

MC+TD3 for Extremal Fewnomial Root Search

Consolidated experimental progress, diagnostic findings, and current direction through 6 September 2026

30/30 stable TD3 runs three-seed 50k confirmation	5/5 criteria passed preregistered transfer test	479/480 safe outcomes all-support robust stage
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Bottom line: MC-trained TD3 reliably transfers to exact root-count improvement. The main limitation is now understood to be systematic proxy mismatch, not Monte Carlo noise. Multi-view scoring has nearly eliminated degradation; the current work is recovering discovery on three over-conservative supports.

Scope

This report consolidates the experiments run from 31 August through 6 September 2026. Training, checkpoint selection, action gating, and candidate selection were label-free. Exact roots were computed only in frozen post-hoc audits unless otherwise noted.

1. Executive summary

The recent campaign moved through three distinct phases: confirm transfer, diagnose failures, and engineer a safer selector. Each phase narrowed the remaining uncertainty and changed the next experiment accordingly.

Date	Question	Result and decision
Aug 31-Sep 1	Does MC-only TD3 transfer reproducibly to exact roots?	Yes. Three seeds, ten supports, 30/30 stable runs, and all five preregistered criteria passed.
Sep 1-2	Are pretraining set size or MC sample count the bottlenecks?	No. 64 and 96 starts tied at 100k; 8x more MC samples barely changed exact correlation.
Sep 2-3	Can long or recurrent rollouts increase discovery safely?	They increase gain, but compound Kac-Rice proxy error on S6/S10.
Sep 3-4	Can diverse MC views reject systematic false positives?	Yes. A 4,096-sample multi-view filter passed blind confirmation on S6 and S10.
Sep 4-5	Does robust MC+TD3 scale to all supports?	Safety scaled: 479/480 nondegrading. Seven of ten supports had positive exact gain.
Sep 5-6	Why did S1/S2/S8 stall?	The explicit no-op action dominated. A no-explicit-no-op multiseed pilot is now running.

What is now established

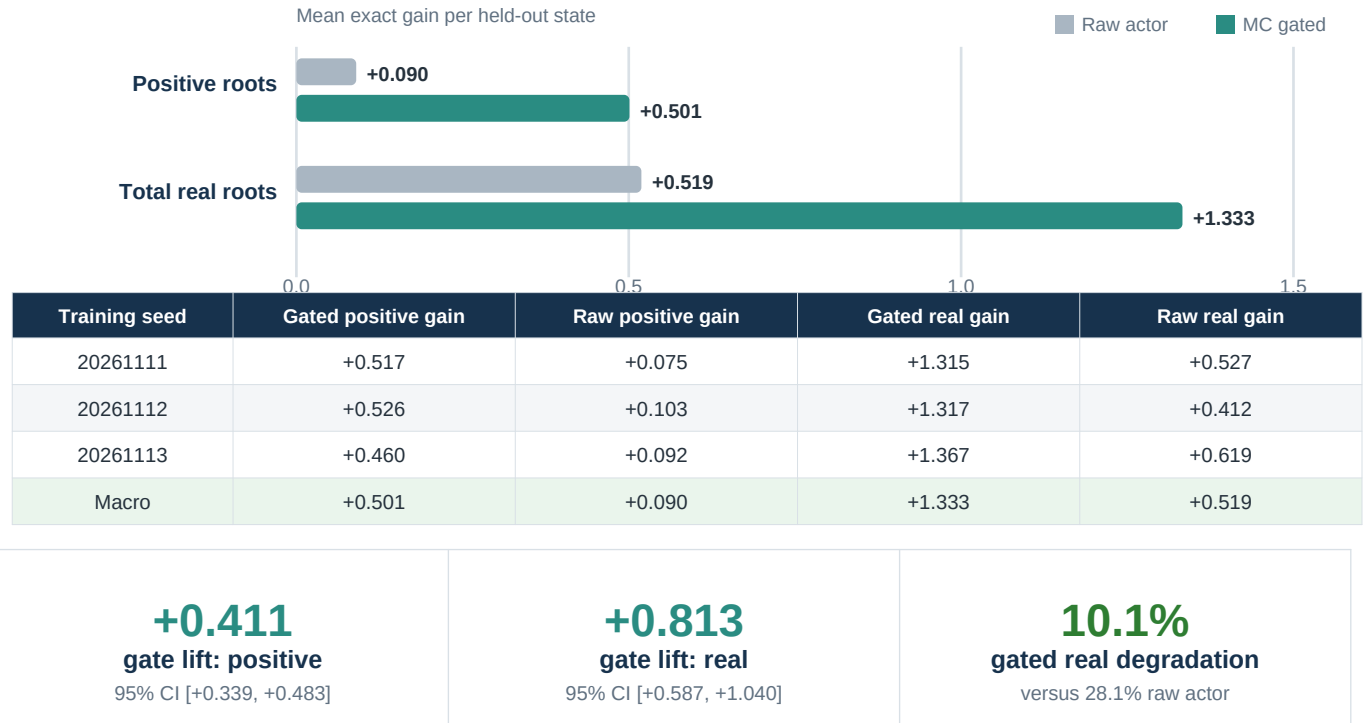
- MC+TD3 proposals are not merely surrogate improvements: the receding MC gate produced positive exact gains across every support and training seed.
- Paired uncertainty estimates improve action-level safety, but additional samples of one Kac-Rice estimator cannot remove structural bias.
- Short, diverse proposals plus multi-view consensus are safer than repeatedly optimizing the same MC proxy.

What remains open

- Whether the no-explicit-no-op variant restores exact discovery on S1, S2, and S8 across multiple seeds.
- How much discovery can be recovered while retaining at least 95% exact real-root nondegradation.

2. Confirmed transfer from MC to exact roots

The 50k confirmatory campaign trained three independent TD3 seeds on each of ten fewnomial supports. Each selected actor was applied in ten receding H3 blocks. Paired Monte Carlo contexts accepted or rejected each block, and all final states were frozen before exact Julia evaluation.



Interpretation

This is the core scientific result. The gate reproducibly converts an MC-trained proposal distribution into improvements in exact total-real and positive-root counts without access to exact labels. All 30 support-seed cells had positive exact gated lower bounds for both endpoints, with zero exact-solver failures.

Preregistered decision: PASS. All five criteria passed, including stability, positive exact gain in every seed, a combined confidence interval excluding zero, and controlled degradation.

3. Efficiency studies and diagnosis

Pretraining budget

A matched 20k screen compared 16, 32, 64, and 96 starting coefficients. All budgets were numerically stable and finished with similar held-out performance. Sixty-four starts became the default compromise. A multiseed 100k extension then showed that 64 and 96 starts were tied at macro held-out H3 gain +0.349.

MC starts	20k held-out H3	20k positive rate	100k macro H3	Decision
16	+0.237	73.3%	-	Promising; slower learning
32	+0.213	73.3%	-	No advantage
64	+0.237	75.7%	+0.349	Default fixed budget
96	+0.231	75.7%	+0.349	Control / adaptive cap

Monte Carlo sample convergence

The audit froze candidate states, varied only the score sample count, and then compared with exact roots. More samples stabilized rankings against the 1,024-sample reference, but did not materially improve agreement with the exact objective.

Quantity	128 samples	1,024 samples	Change
Mean repeat log-score SD	0.066	0.024	Large variance reduction
Endpoint Spearman vs exact roots	0.578	0.580	+0.002
Local-gain Spearman	0.471	0.500	+0.029
Relative scoring cost	1x	8x	+7x

Diagnosis: finite-sample variance is controllable, but the remaining error is predominantly objective mismatch. The efficient policy is 128 samples during training, 512 for ambiguous or shortlisted actions, and 1,024 only for final ranking.

4. Mapping the safety-discovery frontier

The next experiments tested whether longer search, actor diversity, and explicit conservatism could separate gain from degradation. Results below are from independent fresh-state samples, so they define an operating frontier rather than paired treatment effects.

Method	Support	Actions	Mean exact real gain	Nondegradation	Finding
Paired adaptive ensemble	S3	30	+1.292	93.8%	High gain; misses safety
Paired adaptive ensemble	S6	30	+1.250	93.8%	High gain; misses safety
Paired adaptive ensemble	S10	30	+1.917	89.6%	Highest gain; unsafe
Conservative TD3	S3	3	+0.083	93.8%	Very short; weak discovery
Conservative TD3	S6	3	+0.500	100.0%	Safe local policy
Conservative TD3	S10	3	+0.208	97.9%	Safe local policy
Recurrent conservative	S3	30	+1.125	100.0%	Recurrence succeeds
Recurrent conservative	S6	30	+0.583	87.5%	Proxy error compounds
Recurrent conservative	S10	30	+0.042	91.7%	Proxy error compounds

Scientific interpretation

- Thirty-action ensembles demonstrated search reach, including 12 real roots on S10, but did not consistently meet the 95% safety gate.
- A three-action conservative policy moved the operating point toward safety, with 97.2% aggregate nondegradation and +0.264 aggregate mean gain.
- Reusing one actor for ten blocks worked on S3, but systematically amplified false positives on S6/S10. S10 paired MC gain versus exact gain had Spearman -0.036.

Resulting design rule: use recurrence only where validated. On harder supports, obtain discovery from independent actors, restarts, and score views rather than a deeper trajectory under one biased proxy.

5. Multi-view scoring: the safety breakthrough

The robust selector combined three independently trained H3 actors with nine Kac-Rice delta/domain views, eight scrambled-Sobol contexts, and an adaptive 4,096-sample cap. Instead of trusting one low-variance estimate, it required agreement across deliberately different views of the proxy.

Locked validation	Previous dual-LCB	Multi-view consensus	Effect
S6 mean real gain	+0.208	+0.250	Improved
S6 nondegradation	85.4%	93.8%	+8.4 points
S10 mean real gain	+0.500	+0.500	Preserved
S10 nondegradation	95.8%	97.9%	+2.1 points

Fresh blind confirmation

Support	Mean real gain	Nondegradation	Mean positive gain	Absolute / gain Spearman	Primary
S6	+0.458	100.0%	+0.146	+0.808 / +0.641	PASS
S10	+0.375	96.9%	+0.281	+0.837 / +0.512	PASS

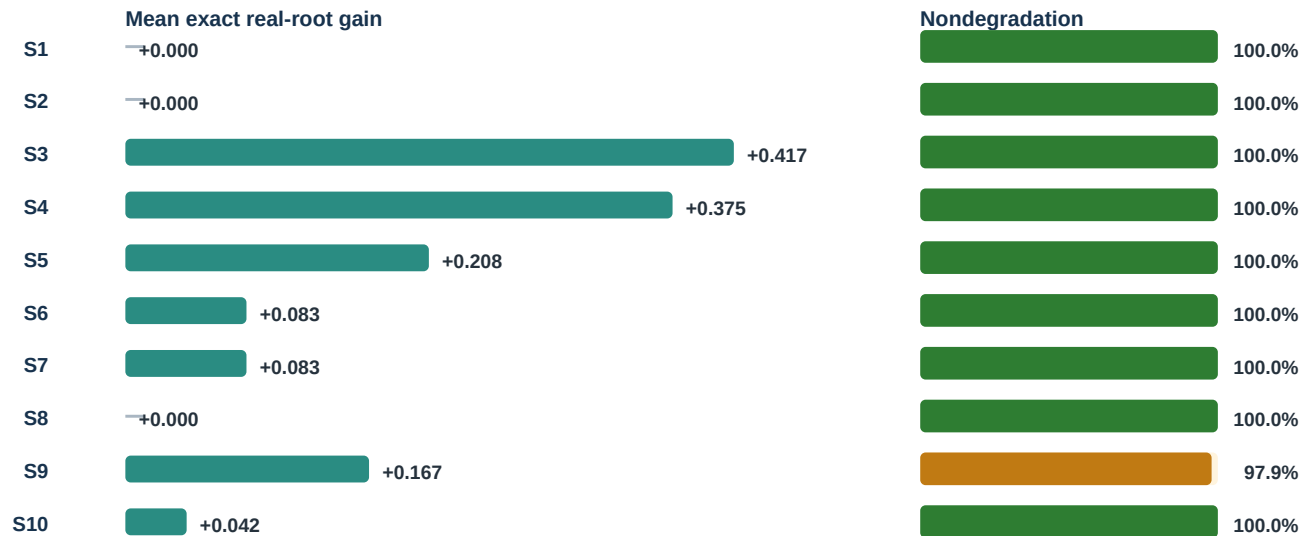
96 fresh starts per support	9 independent views three scales x three deltas	4,096 adaptive sample cap 16,384 was not better
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The 16,384-sample exploratory pass accepted more systematic S6 false positives. This confirms the central diagnosis: view diversity changes the bias profile, whereas additional samples from the same view mainly narrow confidence around the wrong cases.

Primary confirmation: PASS on both S6 and S10. No exact labels were used for training, proposal generation, MC scoring, gating, archive selection, or shortlist selection.

6. All-support robust MC+TD3 stage

The multi-view conservative configuration was then trained from scratch for 20k steps on all ten supports using one health-gate seed. The exact audit covered 480 held-out episode archives, all frozen and hashed before Julia evaluation.



479/480
real-root nondegrading
99.8% overall safety

7/10
supports passed
safe with positive exact gain

+0.138
macro real gain
single-seed 20k stage

Interpretation

Safety generalized exceptionally well: only one of 480 states lost exact real roots, on S9. Discovery did not generalize uniformly. S1, S2, and S8 retained their initial state almost always and had zero mean exact real-root gain. The method therefore reached a highly conservative point on the frontier rather than failing numerically or becoming unsafe.

The stage passed its health objective but not its all-support gain objective. It justifies a targeted exploration ablation, not an indiscriminate longer campaign.

7. Current experiment: removing the explicit no-op

Inspection of the selected all-support policies identified a plausible mechanism for the stalled supports. The actor had an explicit no-op output and selected it in 83% of S1, 85% of S2, and 92% of S8 held-out decisions. This made the system safe, but left almost no opportunity for the external robust gate to evaluate useful moves.

Support	Prior explicit no-op rate	Prior accepted rate	Current completed step	Current H3 MC gain	Current accepted rate	Best robust score
S1	83%	5%	4k / 10k	+0.124	33%	+0.094
S2	85%	3%	5k / 10k	+0.143	44%	+0.118
S8	92%	1%	5k / 10k	+0.117	45%	+0.129

Pilot design

- Target supports: S1, S2, S8. Controls: S3, S4.
- Three training seeds, 10k online steps, 15 jobs total, with three jobs running concurrently.
- The actor must propose a six-dimensional move; the existing paired-LCB rejection gate and episode-incumbent fallback still provide safety.
- The long campaign launches only if pooled exact nondegradation is at least 95%, pooled gain is positive on each target, and at least two of three seeds are positive.

Status at report time: encouraging but interim. The first seed shows positive MC validation on all three stalled supports and substantially more accepted moves. These are not yet exact-root results; the frozen multiseed Julia audit remains decisive.

Decision framework after the pilot

Outcome	Action
Targets and controls pass	Launch the longer multiseed no-explicit-no-op campaign.
Targets gain but safety falls	Retune the external LCB threshold or reduce action scale; do not restore the actor no-op.
MC improves but exact gain remains zero	Treat the supports as objective-mismatch limited and add actor/view diversity rather than training length.
Controls regress	Reject the ablation as a general default and keep it support-specific.

8. Conclusions and recommended next steps

Conclusions

- The principal thesis claim is now supported: an MC-only TD3 pipeline can produce statistically reliable improvement in exact fewnomial root counts.
- The receding MC gate is essential. It increased exact gain and reduced degradation in the preregistered three-seed study.
- Larger pretraining and scoring budgets have diminishing returns because the hard errors are systematic, not merely noisy.
- Long trajectories expose proxy exploitation. Independent H3 actors and multi-view scoring are the more reliable route to discovery.
- The all-support robust stage achieved near-perfect safety. The remaining engineering question is how to restore proposal diversity on S1/S2/S8.

Recommended sequence

Priority	Experiment / analysis	Decision value
1	Finish the 15-run no-explicit-no-op pilot and its frozen exact audit.	Tests the current bottleneck directly.
2	If the gate passes, extend only the successful support/configuration cells to longer training.	Avoids another expensive all-support campaign before evidence warrants it.
3	Keep 128 training samples, adaptive 512 rescores, and 1,024 final ranking; reserve 4,096 multi-view scoring for robust selection.	Maintains the best observed cost/safety balance.
4	Use recurrent 30-action search on S3 only; use independent actors/restarts on S6/S10 and any other mismatch-limited support.	Prevents repeated exploitation of one proxy direction.
5	Report safety and discovery separately in all future tables.	Makes frontier movement visible and prevents a safe no-op policy from appearing successful.

Overall assessment: strong progress. The program has moved beyond proof of concept to a controlled, blind-audit workflow with a quantified safety-discovery tradeoff and a focused next experiment.

Appendix A. Evaluation conventions and evidence

Term	Meaning in this report
H3	A three-action actor rollout block.
MC gain	Change in the Kac-Rice Monte Carlo proxy score on held-out contexts.
Exact real gain	Change in the exact number of total real roots computed by Julia.
Exact positive gain	Change in the exact number of positive roots.
Nondegradation	Fraction of held-out states whose selected archive does not lose exact real roots.
LCB	Lower confidence bound used to reject proposals whose estimated gain is not sufficiently positive.
Blind audit	Candidate queue and hashes frozen before exact labels are computed.

Primary local evidence files

- latent_td3_repro3seed50k/FINAL_REPORT.md
- latent_td3_pretrain_budget_ablation20k/PRETRAIN_BUDGET_ABLATION_REPORT.md
- latent_td3_pretrain_budget_extension100k/analysis/REPORT.md
- kac_rice_mc_sample_convergence_s3_s6_s10/REPORT.md
- paired_adaptive_ensemble30_confirmed_fresh48_s3_s6_s10/posthoc_mc1024_exact_audit/REPORT.md
- latent_td3_conservative_pilot12k_s3_s6_s10/posthoc_mc1024_exact_audit/COMPARISON_REPORT.md
- recurrent_conservative_td3_30action_fresh48_s3_s6_s10/posthoc_mc1024_exact_audit/COMPARISON_REPORT.md
- mc_multiview_robustness_s6_s10_one_shot336/analysis/REPORT.md
- robust_mc_blind_confirmation4096_gaussian96_s6_s10/posthoc_blind_julia_audit/REPORT.md
- robust_mc_td3_all10_stage1_20k/posthoc_blind_julia_audit/REPORT.md
- robust_mc_td3_no_explicit_noop_pilot_10k/pilot_manifest.json and current status files

Evidence note: comparisons across separately generated fresh-state sets are descriptive and are not treated as paired causal estimates. Interim no-explicit-no-op numbers are MC validation metrics at the latest completed checkpoints available when this report was generated.